



C2IMPRESS

**D2.3: SoS4MHRIN Platform for C2IMPRESS –
methods compendium and report v2**

WP2

3 February, 2025

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2. Acronyms

Acronym	Description
C2IMPRESS	Co-Creative Improved Understanding and Awareness of Multi-Hazard Risks for Disaster Resilient Society
GA	Grant Agreement
ESDI	Earth System Dynamic Intelligence
IPAI	Information Physical Artificial Intelligence
MET	Synergistic Manifolds Unip. Lda. (Meteoceanics)
SoS4MHRIN	System-of-Systems for Multihazard Risk Intelligence Networks
QITES	Quantum Information Technologies in the Earth Sciences
AIPSI	Augmented Information Physical System Intelligence
IPQuTI	Information Physical Quantum Technological Intelligence



3. Executive Summary

The present report pertains the scientific and technical operationalisation of the theoretical advances and methodological developments undergone in fulfilment of WP2 Tasks 2.1, 2.2 and 2.3 towards a System-of-Systems for Multi-Hazard Risk Intelligence Networks (SoS4MHRIN), along with an overview of key features and applications to the broader C2IMPRESS project mission. Having thoroughly detailed the fundamental formal theoretical and methodological developments in the related prior deliverable D2.1, the present contribution aims towards a more succinct, practical account on the operational model development and system implementation, building from the recent underlying works developed at MET, namely by Perdigão (2024) and Perdigão and Hall (2023, 2024).

In practice, the report covers three main steps to SoS4MHRIN operationalisation:

Firstly, translating the theoretical framework reported in D2.1 into operational form leveraging MET's Augmented Information Physical System Intelligence AIPSI (Perdigão and Hall 2023) and Information Physical Quantum Technological Intelligence IPQuTI (Perdigão and Hall 2024). These entail core advances in nonlinear natural-based information physics, along with advances in far-from-equilibrium thermodynamics and evolutionary cognition. Providing a native operational environment to capture, process and relay information in an interoperable manner across classical, quantum and post-quantum platforms, in a unified information physical manner: physically consistent across scales from infinitesimal fluctuations in quantum gravitation all the way up to large-scale astronomical processes in thermodynamic cosmology, in material fulfilment of Perdigão (2017) theoretical constructs.

Secondly, outlining the novel operational model structures, along with the associated datasets, underlying principles and governing mechanisms at play in the complex multiscale multidomain Earth System Dynamics, including the emergence, coevolution and synergy among key mechanisms at play.

Thirdly, leveraging the theoretical developments in D2.1 and the operational developments herein to implement SoS4MHRIN in the operational form, producing visual outcomes to be included in the user interface, which is also outlined ahead of its final completion and demonstration to the end users.

The operationalisation of SoS4MHRIN further puts into evidence the platform purpose, capabilities and functionalities in articulation with the overall project. This includes, but is not limited to, the CSA contexts, given the cross-cutting portability and broad generality of the theoretical, methodological and operational framework across spatiotemporal scales, to be further explored in pan-European and global scale simulations of key individual, compound, synergistic and coevolutionary processes and interactions underlying the overall spatiotemporal multi-hazard and Earth system dynamic nexus.



4. Disclaimer: Unforeseen Issues and Constraints

An unforeseen major incident severely harming the MET team in January 2025 has resulted in a medical emergency that adversely impacted the completion of this deliverable, delaying its submission.

In terms of constraints, the presently reported results reflect the work conducted in WP2 Tasks 2.1, 2.2 and 2.3 until Month 28. Task 2.2 continued until the end of the month even as this deliverable D2.3 was being completed. And Task 2.3 still unfolds until Month 30, thereby bringing additional results that will be reflected onto the final form of SoS4MHRIN. In other words, while the present deliverable is final, there are further activities within the regular project schedule that shall further enhance SoS4MHRIN and its user experience ahead of its milestone demonstration on M32.

Notwithstanding these unforeseen emergency circumstances and constraints, the deliverable still endeavours to fulfil the pledged goals of presenting the SoS4MHRIN platform's functions, methods and user benefits, along with the output that are made available to the CSAs.



1. Introduction

1.1 General Motivation

The Earth System can be regarded as a multiscale multidomain far-from-equilibrium complex coevolutionary system, drawing its structural-functional coherence and coevolution from a delicate duality between order and chaos, between the emergence of dissipative structures amidst entropy-producing chaotic behaviour (Perdigão 2021). The complex system dynamics shaping the behaviour of such diverse phenomena as weather regimes, hydrologic flows, ecosystem dynamics, along with the emergence of highly dissipative destructive disasters from floods to wildfires, calls for system dynamic frameworks reflecting the more realistic nature of our critically changing planet. One that inherently operates in highly nonlinear, non-ergodic hyper-chaotic motion, symmetry breaking and unprecedented emergence. One that goes beyond the ill-posed assumption of symmetries, recurring patterns and stable equilibria pursued by state-of-art analytics and modelling approaches ranging from physical models to machine learning, deep learning and artificial intelligence (AI). One that goes beyond prior data and models to provide novel synergistic dynamic mechanisms at the onset of nonrecurrent structural-functional coevolution and system dynamic innovation (Perdigão 2018).

Traditional process-based and data-based approaches are rooted on notions of system invariants, patterns and symmetries. Therefore, they are only moderately appropriate in highly idealised settings, such as specific transient metastable conditions where some recurrent behaviours arise, e.g. prevalent weather regimes mistaken for patterns when in fact they are temporary loops, and the ill-assumption of a stable climate where such patterns could be drawn from datasets and then be used to inform model architectures ranging from physically informed differential equations all the way to data-informed generative artificial intelligence frameworks. The problem being that the climate is not stable, hence the learned regimes in the training stage are no longer valid in future system dynamics.

In order to overcome these limitations, MET had developed Earth System Dynamic Intelligence (Perdigão 2021) and associated theoretical and methodological developments also outlined in D2.1, along with methodological upgrades such as Augmented Information Physical Systems Intelligence (Perdigão and Hall 2023), Information Physical Quantum Technological Intelligence (Perdigão and Hall 2024), synergistically articulating emerging pathways in information physics and causality (Perdigão et al 2020, Hall and Perdigão 2021), generalized categorical algebraic constructs (Martins-Ferreira and Perdigão 2024), along with neuro-quantum dynamic geometry and topology (Perdigão 2024).



Figure 1.1 – Schematic depiction of the proprietary Synergistic Manifolds System-of-Systems Intelligence and associated methodological constellation developed and operated at Synergistic Manifolds - MET (Perdigão and Hall 2024).

The mathematical physical underpinnings are understandably too formally involved and complex to be straightforwardly discerned by applied researchers, practitioners and end users, even in high-level technical areas. Therefore, the operationalisation of SoS4MHRIN in C2IMPRESS required a delicate effort to keep simplicity and clarity amidst the mathematical depth of the underlying methodological developments, while keeping physical consistency and focusing on operational usability, so that it could provide end-users, irrespective of their technical background, a way to benefit from the results of our research and development in the form of intuitive, interpretable and actionable outcomes for a diversity of workflows ranging from informing predictions, scenario building and decision support.

The present deliverable thus aims to provide a balanced effort where a brief overview of our scientific and technical advances is reported in operational model development, along with usable outcomes at CSA and broader spatiotemporal levels, thereby illustrating the portability and generality of our solution to serve not only specific case studies in C2IMPRESS but rather beyond, either in Europe or the broader global context. Thereby providing value at and beyond project level, so that it can help communities across the world to better understand, assess, predict and act upon our broad findings.

2. Operational Model Conceptualisation

2.1. Structural canvas and spatiotemporal observables

Our operational model development begins with the establishment of a structural canvas upon which to build the systemic edifice embodying the processes and interactions at play in the dynamics. In practice, the structural canvas is hereby represented by an operator space S^b endowed with inner product, differential-geometric structure and higher-order nonlocal links up to infinite b , in order to encompass and articulate the overall system dynamics from local to global, in space and time. In doing so, S^b acts as global spatiotemporal generalisation to Hilbert-Sobolev spaces (Courant and Hilbert 1953, Sobolev 1938, Sobolev 1963), admitting infinite spatiotemporal dimensions, local and nonlocal integro-differential operators defined up to infinite-order integer derivatives and infinitesimal-order fractional integro-derivatives for unbounded non-local interactivity, building from Perdigão (2017). Fractional integro-differentiability enables differential operators to be formally defined everywhere in the domain with neither locality nor smoothness requirements, unlike traditional derivatives. This is particularly useful when evaluating fluid flows over critical transitions e.g. across thermodynamic discontinuities in weather fronts, and along intermittent processes such as rainfall or flood events.

With the canvas at hand, a set of observables is then defined: $X_{s,t}$. These denote the set of physical variables or fields that can be observed, measured and recorded in spatiotemporal datasets, where s represents the spatial coordinates and t the time, in as many dimensions or scales as needed. While multiple spatial dimensions has a straightforward geometric intuition, multiple time dimensions can be understood in a spectral manner, as discernible timescales or ‘clocks’. This proves useful in such settings as polispectral weather forecasting, with multiscale slow-fast-faster cascades such as at the transition from wavelike synoptic flows to turbulent fluxes traditionally elusive to classical mechanics.

Upon observation or measurement, the observables are mapped into observations or measurements, $Y_{s,t}$, corresponding to the practical quantities the dynamics of which are recorded into datasets for further processing. These are then stored as spatiotemporal functions embedded or projected into a subspace S^b , subject to physical and operational constraints associated to eventual limits in the dimensionality d and dynamic connectivity c of the system, resulting in finite $b'=(d,c)$.

When the mapping is performed by system dynamic embedding (our choice), it entails an information-invariant or lossless transformation. When it is performed via projection, then it entails information loss through the contraction properties of the associated inner product. Whether an embedding or a projection takes place depends on whether the observation/measurement is performed in a



conservative or dissipative manner, e.g. whether or not the gauging instrument captures and preserves the physical integrity of the observable into producing the observation/measurement. The latter then becomes the mathematical object which will participate in the model structure to be built.

2.2. Spatiotemporal Interaction Structures

With canvas and objects at hand, the system now requires morphisms, or transformations, representing the formal ways in which the objects interact, e.g. by exchanging energy, mass, or more generally information in its more fundamental thermodynamic information-theoretic algebraic form. The overall complexity of the system will then be critically defined not in terms of the cardinality and dimensionality of the objects, but rather in terms of the entanglement level of the interactions.

In order to formulate interactions, it is important to note that linear or nonlinear correlation measures do not suffice, since they provide aggregate statistical relations without considerations of dynamic nature. We will also be interested in the information content stemming from higher-order nonlinear dynamical interactions and coevolutionary interplay, including the role of microphysical codependencies in shaping the macrophysical information metrics, as derived in Perdigão (2018B).

In the scope of the present work, we capture the dynamics at play between processes through Dynamic Interaction Analysis (DIA) following Perdigão (2018). Essentially, the procedure consists of evaluating local and nonlocal linear and nonlinear dynamical sensitivities among processes, i.e. their dynamic interactions, along with their time evolution and spatial variability. This will be a fundamental building block for our operational dynamic model design.

Mathematically, the *Multiscale Dynamic Interaction (MDI) of rational k^{th} order among n spatiotemporal multiscale multivariate processes $M_1, \dots, M_n$* , generalizes Perdigão (2018) into:

$$\mathcal{D}^k(M_1, \dots, M_n) \equiv \nabla_{(M_1, \dots, M_n)}^k (\dot{M}_1, \dots, \dot{M}_n)$$

where the rational-valued k enables local (integer order) and nonlocal (fractional order) dependencies among multiple multivariate processes, be it in the scope of multivariate classical systems or multipartite quantum entanglement for the quantum-ready implementation of the computations. Each multivariate process can be either classically local or a distributed quantity such as a wavefunction or a more general nonlocal structure in our generalised operator spaces. The application of fractional-order interaction diagnostics results in capturing nonlocal codependencies without requiring local differentiability or even continuity, which is useful in intermittent processes.

In particular, the local first order dynamic interaction between two processes living in a vector space becomes equivalent to the Jacobian matrix of their coupled dynamical system. Another particular case entails the Koopman matrix of cross-operators in a Hilbert space, expressing the functional representation of codependencies among variables in a multilinear framework.

While particular approaches entailing linear or multilinear approximations would suffice in some applications where local equilibrium is still prevalent (e.g. smooth flows), that is not the case in multiscale multidomain Earth System Dynamics. Here, the extreme nonlinearity and criticality of cascading, compound, synergistic and coevolutionary processes requires a full-fledged system dynamic operator framework without linearisation concessions. Nonlinearity manifests even at the most subtle, local, particular settings, such as microphysical turbulence, anomalous diffusion in fluids, and other processes that, whilst microphysical, cascade all the way up to the measurable macroscales making a significant difference into how phenomena unfold, how extreme events emerge, interact and impact the system across spatiotemporal scales.

To capture such extra features of system dynamic complexity, elusive to traditional thermodynamics and information-theoretic frameworks rooted on Boltzmann-Gibbs-Shannon entropies, Perdigão



(2018) has introduced novel information physical metrics generalising classical metrics of entropy, synergy and redundancy to capture also these emerging multiscale features, including where the role of the microphysical complexity is also taken into account. This is also explained in deliverable D2.1.

In an operational system dynamic modelling setting, the nonlinear statistical mechanic representation underlying information theoretic metrics is further complemented with MDI as a kinematic-geometric expression capturing local and nonlocal spatiotemporal dynamics in real time.

By evaluating all self and cross sensitivities of various orders within the system, the measure in the aforementioned equation captures linear and nonlinear dynamic interactions required to characterise the dynamic feedback structure of the system. The rational generalisation further enables nonlocal spatiotemporal codependencies to be harnessed, including in terms of spatiotemporal teleconnections, persistence and spatially distributed long-term memory.

In practice, we are interested in unveiling spatial and temporal structures of the dynamic interactions among processes, e.g. the time-integrated geographical distribution of an ocean-atmospheric flow or the spatially integrated evolution of a local, regional or global geophysical quantity e.g. a climate index. More than statistical aggregates entailing information loss, we will be interested in information-preserving integration processes, hence in a differential-geometric way.

For that purpose, we consider the general spatial structure of the interaction of k^{th} order between processes $\mathbf{M}_i$, $i \in \{1, \dots, n\}$ captured over a time period T . This structure can then be expressed by the spatial submanifold of their dynamic interactions, through an information-preserving coevolutionary system dynamic embedding of the higher-order manifold into the spatial submanifold. This prevents the information loss associated to classical integration by averaging, aggregation, contraction or projection that would alienate information about temporal variability. Analogously for time, the temporal structure captured over a certain spatial region S_r is given by the coevolutionary spatiotemporal system dynamic embedding into the temporal dimension, from which the spatial information can still be unfolded via the spatiotemporal coevolution manifold (Perdigão 2017), comprising relative celerities among the manifold processes, harnessing the temporal legacy engraved in spatial structures, and the spatial legacy engraved in temporal variability.

The operational mathematical physical underpinnings are introduced and laid out in more detail in Information Physical Quantum Technological Intelligence (Perdigão and Hall 2024), where the local and nonlocal fractional-geometric interactions are evaluated among Earth System Dynamic processes contributing to the emergence of multi-hazard system dynamic behaviours. The transition from local to global is made through our generalised nonlocal differential geometric composition linking the system dynamics at local chart level towards losslessly integrated global maps, as a generalisation of manifold construction in differential geometry. While in traditional differential geometry and topology the construct is rooted on local differentials, in our generalised case we build operator gates entailing nonlocal integro-differential operators to enable such system construction without the stringent requirements for local smoothness. This proves particularly crucial in critical situations like intermittent processes such as precipitation events, floods, droughts, wildfires and their interactions.

2.3. Spatiotemporal Model Design

The backbone of system dynamic model design resides in the interactions. Irrespective of the number of variables involved in a system, complexity will arise not from these, but rather from the intricacy and generative power of the interactions in producing the rich diversity of system dynamic features aiming to represent key aspects of interest in the real-world dynamics under analysis. Therefore, the model design will trivially invoke MDI to capture multiscale multidomain dynamic interactions among the processes taking part in the dynamics. If the set of processes forms a dictionary, the MDI provides the grammar through which the processes are articulated with each other to produce the story being



told by the model about the phenomenon of interest, e.g. how precipitation unfolds as a function of underlying geophysical mechanisms engraved in geopotential height fields driving the dynamics.

In order to maximise efficiency of the model structure, minimising redundancies among processes so that the interactions will themselves not feature redundant exchanges, we perform MET's dynamic source analysis explained in D2.1 (a part of our information physical system dynamic intelligence). This nonlinearly disentangles the raw spatiotemporal geophysical fields such as multiscale multilayer geopotential and brings out the minimum number of fundamental non-redundant operators that synergistically articulate to produce the observable coevolutionary behaviour in ocean-atmospheric circulation regimes, including the nexus involving winds, ocean currents, phase transitions and mass fluxes (e.g. evaporation, precipitation), along with thermal and thermohaline (ocean) processes.

While the observed physical phenomena does entail a plethora of interacting processes, leading to the apparent notion that there is nothing separable underneath, Perdigão (2018) had shown that underneath a coevolutionary system a clean synergistic set of dynamic sources could be unfolded. This eliminates the redundancy in model design whilst representing the system dynamic not in terms of observations but in terms of more fundamental physical observables, in a core set of principles. For this reason, the system dynamic intelligence does not entail a brute force modelling over a myriad of redundant variables, but rather in terms of a handful of fundamental physical mechanisms. This is also in line with core foundations in geophysical sciences, wherein even the most elaborate weather models all come down to a small set of primitive equations governing momentum, thermodynamic and moisture fluxes. Yet unlike classical weather models, the governing equations are not merely macrophysical (requiring parametrisations for unresolved subgrid scales and erroneously assuming local homogeneity and equilibrium), but rather multiscale taking into account the microphysical complexity as explained above.

By not assuming classical fluid properties such as homogeneity and local equilibrium (violated in extreme storms, fire-weather systems, interacting hazards just to name a few), the new generation of operational models can aim to better capture such layers of complexity that are increasingly more prevalent and pertinent in the face of the anomalous thermodynamic challenges facing our planet.

Therefore, having retrieved dynamic sources $\mathbf{X}$ (e.g. fundamental geophysical observables) underneath an observed coevolutionary dynamical system (e.g. geopotential height fields), we establish a framework for the simulation of predictand processes $\mathbf{W}$ (e.g. precipitation, streamflow, temperature, momentum). This can be addressed as an apparently simple general nonlinear differential model, where the processes of interest $\mathbf{W}$ depend on their interplay with sources $\mathbf{X}$:

$$\dot{\mathbf{W}} = \sum_{k=0}^q k!^{-1} [\mathcal{D}^k(\mathbf{W}, \mathbf{X})]_{\mathcal{R}} \mathbf{X}^k + \mathcal{O}(\mathbf{X}^{q+1})$$

where each variable is a spatiotemporal functional object in H^b (think of it as the operator algebra equivalent to a field, a wavefunction or a distribution), the k^{th} order powers $\mathbf{X}^k$ account for nonlinear interactions of k^{th} order within each component of $\mathbf{X}$, $\mathcal{R}$ denotes evaluation at the reference manifold, and $\mathcal{O}(\mathbf{X}^{q+1})$ refers to the residual of higher-order dynamics beyond the analytical threshold at order q . On non-critical cases, a truncation can be set manually to limit the system representation up to a certain order of interactions. However, in general we let the system interact with a boundless q , so that the truncation only takes place when the higher-order residual is of null measure, indicating the absence of higher-order interactions unaccounted for. In doing so, information loss is prevented.

The above system provides a general model structure template or recipe to address the dynamical evolution of any system in terms of a functional basis comprised of underlying independent controls. **The determination of a functional basis in a dynamical system is an important step to its optimal modelling, as it ensures that the system is expressed in terms of the smallest number of dimensions**

without loss of generality. By taking into account not only linear elasticities (D^k for $k = 1$) but also nonlinear codependencies (D^k for $k > 1$) and nonlocal dependencies (D^k for k fractional), this system can then address changes in dynamical tendencies in terms of nonlinearly interacting processes.

3. Operational Model Use Cases

Having conceptualised the generic operational model design structure, example use cases are now considered pertaining some geophysical processes and interactions at play in Earth System dynamics. **The outputs of said simulations will be set for inclusion in the end-user platform, where CSA partners and other interested parties shall access, visualise and download the corresponding outputs, and consult the associated explanations for drawing informative insights on the dynamics at play. Thereby improving the awareness and understanding of these dynamics in an actionable operational way.** An overview of the features, functionalities and examples of the visual interface through which the user shall communicate with SoS4MHRIN is presented in section 5 further below.

3.1. Characterisation of Geophysical Observables and Datasets

One of the most relevant geophysical observables that embodies the dynamical imprint of multiscale interacting processes is the spatiotemporal Geopotential Height fields at pressure levels p , ϕ_p .

Fundamentally, ϕ_p provides information on the geospatial dynamics of atmospheric waves at the pressure level p to which ϕ_p refers, e.g. $p = 500$ hPa. High and low pressure centres can be detected in geopotential terms as high and low heights of a given pressure level. In particular, the large-scale fields provide information on low-frequency atmospheric waves at the planetary scale.

In their natural form, ϕ_p contains not only information on the state of the atmosphere at the pressure level p but also on how it is being affected by underlying forcing mechanisms taking part in its dynamics. In fact, embedded at the fields for each pressure level lie dynamical footprints of processes at other levels, through vertical interactions that ultimately bring in features stemming from other geophysical domains. These include oceanic and land-surface processes at large scales, resulting in geopotential heights actually responding to the difference between the state of the atmosphere over large water and land bodies such as oceans and continents, even considering mid-upper pressure levels such as 500 hPa. Higher pressure levels (corresponding to lower vertical levels) will naturally have a finer sensitivity to the surface properties, at the expense of getting lost in the details and losing the grasp on the bigger picture. Therefore, our interest spans the whole vertical column of levels.

The geophysical mechanisms captured in the geopotential height fields are in turn at the source of the atmospheric circulation regimes that ultimately shape the weather, including moisture advection and thermodynamics, underlying the precipitation dynamics we intend to understand. Comprehensive, introductory treatments on associated atmospheric physical processes are found at [Salby 2012] and [Holton 2004].

One of the most reliable and popular datasets with geopotential height fields stems from the NCEPNCAR Reanalysis Project [Kalnay et al. 1996]. Therefore, it is one of the geopotential datasets of choice for our analysis. Here, our dynamic model design methodology is implemented onto NCEP-NCAR Reanalysis data on the multilevel Geopotential Height fields for a subset of the Northern Hemisphere spanning $[30,70]$ degrees of latitude North and $[-80,40]$ degrees East of longitude. This area covers most of Europe, the Mediterranean and the North Atlantic. Other variables retrieved from the same reanalysis suite include Temperature, horizontal and vertical winds, relative humidity, from which the remainder variables are derived including relative and absolute vorticity (curl of the velocity field) along with a representative set of thermodynamic potentials relevant to the verification of adequacy in the system dynamic modelling of highly nonlinear dynamic convection and turbulence.



Another key source of comprehensive geophysical information for modelling verification was the Copernicus Climate Change Service, Climate Data Store (2024), Hersbach (2024). For example, upon retrieving the full fledged ERA5 post-processed daily-statistics on pressure levels from 1940 to present, the following variables were drawn: Horizontal divergence of velocity; Fraction of cloud cover; Geopotential; Ozone mass mixing ratio; Potential vorticity; relative humidity; specific cloud ice water content; specific cloud liquid water content; specific humidity; specific rain water content; specific snow water content; Temperature; Eastward Component of horizontal Wind (U-wind); Northward component of horizontal wind (V-Wind); vertical velocity (W-wind); relative vorticity. While pervasive redundancy exists among those variables, our dynamic source analysis retrieved the underlying low-dimensional physical core, which in essence is found embedded into the multiscale fluctuations of the key thermodynamic potentials for the Earth system, namely the geopotential spatiotemporal fields.

3.2. Retrieving Model Building Blocks from Spatiotemporal Datasets

Geophysical fluid datasets provide discrete-sampled records of the mostly continuous spatiotemporal fields through which the underlying fluid dynamics unfolds in nature. Therefore, a key challenge in geophysical fluid analysis and modelling pertains the reconstruction of said spatiotemporal continuum from its discrete sampling records. Traditionally, statistical interpolations would be performed over space and time under stringent assumptions of local equilibrium, homogeneity and smoothness of the natural fields. In doing so, local perturbations such as microphysical turbulence seeding multiscale turbulent cascades would remain elusive, thereby hindering the ability to capture the onset of unstable dynamics leading to critical transitions and extremes. This is one of the reasons why traditional models in meteorology, oceanography and hydrology struggle to capture such critical dynamics, often underestimating the emergence of highly localised pattern-breaking extreme events.

In order to overcome these limitations, we operationalise MET's Information Physical System Dynamic Intelligence explained in the previous Deliverable D2.1, also benefitting from the generalised fluid dynamical systems framework first introduced in Perdigão (2017). In doing so, we can position ourselves in a more general fluid dynamical regime allowing for local instabilities, local inhomogeneities, nonlocal dependencies and fractalities (instead of local equilibrium, local homogeneity and smoothness from state-of-art approaches still using Navier-Stokes equations).

This allows for local anomalies, codependencies and coevolutionary interplay to be captured, along with their scaling effect all the way up to the macrophysical level where typically fluid mechanical and thermodynamical quantities are evaluated. The role of microphysics in shaping macrophysical quantities can also be understood and mathematised over data records with the generalised information metrics of polyadic entropy, synergy and redundancy under microphysical codependence introduced in Perdigão (2018B) and also explained in D2.1. Among manifold applications, in our operational system dynamic modelling there is relevance to critical processes as anomalous diffusion, nonlocal entanglement and multifractal scaling cascades in fluids and fluidistic complex systems.

In the present context, a set of spatiotemporal geophysical fields is unveiled from the spatiotemporal geophysical datasets characterised in the previous section leveraging these aforementioned methodologies explained in D2.1, under the physical consistency and nonlinear disambiguation constraints also explained therein, namely in terms of preferential linear and nonlinear tendencies and flow paths in far from equilibrium thermodynamics. Operationally, that entails a fluid dynamical systems reconstruction, unfolding the embedding through which the datasets encode the underlying system dynamics at play, and then performing dynamic source analysis (also explained in D2.1) in a synergistic non-redundant manner minimising the number of source functions whilst retaining the ability to represent the system-level dimensionality and comprehensive spatiotemporal dynamics.

Given the high dimensionality of spatiotemporal geophysical datasets and even more so of retrieved spatiotemporal geophysical fields, dynamic source analysis is essential to eliminate redundancies and



express high dimensional system dynamics into lower dimensional mathematical structure in a lossless manner. The result then emerges in lossless operator form (intuitively akin to a vector file), rather than in lossy grid form (akin to a raster file). Continuous media can then be losslessly expressed as a discrete articulation of nonlocal spatiotemporal field operators instead of a traditional grid.

In several working areas such as image processing or analytical chemistry a polispectral analysis typically suffices to provide such functional representation, e.g. in hyper-spherical harmonics and polyspectral functionals. For example, recurrent trivial systems such as mid-latitude synoptic weather regimes, e.g. winter frontal systems, traditional physical and AI spectral models fare well enough.

However, in the climate system there is an abundance of highly nonlinear non-recurrent non-harmonic processes and interactions at play that justified the development of our methods and tools.

The unveiled source operators, here spatiotemporal geophysical dynamic sources, then provide the building blocks for the operational system dynamic models.

3.3. Multiscale Meteoceanic Circulation Model – A simple example

Classical operational Ocean-Atmospheric circulation models entail high-dimensional model structures either in spectral or in grid form. The spectral high-dimensionality results from the broad dense spectrum associated to non-periodic dynamics at play, i.e. in a virtually infinite number of spatial and temporal harmonics. However, we position ourselves in our information physical nonlinear dynamical systems perspective, enabling such spectra to be aptly compressed into a small set of non-redundant non-periodic modes with full generative potential, i.e. the dynamic sources of the overall system. For that reason, high-order dynamics perceived as involving a large number of processes and interactions can be stylised and conceptualised into a losslessly compacted synergistic dynamical system. The details of the underlying dynamic source analysis have been explained in Deliverable D2.1.

The gridded high-dimensionality results from rasterizing a low-order functional system into a discrete spatiotemporal domain, where most of the information is redundant due to being inherently from the same dynamical system. This is overcome by taking nonlocal spatiotemporal operators instead of pointwise quantities, and formulating the system dynamics through operational gates mediating the interactions among said operators.

As an example, we generalize a low-dimensionality Lorenzian atmospheric system to a nonlocal thermally forced dynamical system of spatially distributed quantities reflecting the teleconnected nature of our changing world, and further forced time-dependent thermodynamic potential regime linking local and event-scale weather dynamics with long-term climate system dynamics leading to structural change and associated stability properties in the system:

$$\begin{aligned}\dot{W} &= -r_{dd}W - (\Phi^2 + \Lambda^2) + \mathbf{e}_\Phi \cdot \nabla_\Phi \Theta \\ \dot{\Phi} &= -\Phi + W\Phi - r_{ta}W\Lambda + \mathbf{e}_\Lambda \cdot \nabla_\Lambda \Theta \\ \dot{\Lambda} &= -\Lambda + W\Lambda + r_{ta}W\Phi + \mathbf{e}_W \cdot \nabla_W \Theta \\ \dot{\Theta} &= r_{gh}\Theta - r_{su}\Theta^{k(W^2+\Phi^2+\Lambda^2)}\end{aligned}$$

As far as variables are concerned, W represents the intensity of the mean zonal flow, taking into account the intensity of the middle-latitude westerly wind current and the cross-latitude temperature gradient in each hemisphere; Φ and Λ represent the strength of the cosine and sine phases of a chain of vortices superposed on the zonal flow transporting the heat poleward in a turbulent manner.

The thermal forcings represent the symmetric cross-latitude solar heating contrast and the asymmetric heating contrast between oceans and continents, varying with longitude. Θ represents the changing global temperature, which has a positive or negative feedback depending on the ratio r_{gh} between heat gains (e.g. by greenhouse) and heat losses r_{su} (dissipative boosting of the dynamics).



The nonlinear term with exponent $k(W^2 + \Phi^2 + \Lambda^2)$ energy limits the fast growth of Φ unstable modes, but unlike canonic bistable systems here it is not necessarily cubic. Instead, it is more akin to Perdigão and Blöschl 2014, with a non-necessarily fixed exponent. This will depend on the energy of the whole system via a proportionality factor k . For a given positive k , the higher the energy of the classical Lorenzian atmospheric circulation dynamical system, the higher the loss of Φ (i.e. the more thermal "fuel" it needs to give to keep the system going -- if it does not, then the system dynamics fade and with it the temperature increases more as heat is being stored instead of being spent as work).

3.4. Dynamic Modelling of Evolving Precipitation Distributions

The dynamical simulation model introduced in section 2.3 as a general simulation framework, is now applied to the simulation of evolving Precipitation distributions from the Dynamic Sources of the Geophysical fields. Taking $\mathbf{Z} = \mathbf{P}$ in said equation and determining the threshold beyond which no non-null measure dynamics emerges to be $q=5$ bring us to a practical model form:

$$\dot{\mathbf{P}} = \sum_{k=0}^5 k!^{-1} [\mathcal{N}_T^k(\mathbf{X}, \mathbf{P})_{\text{eff}}]_{\mathcal{R}} \mathbf{X}^k + \mathcal{O}(\mathbf{X}^6)$$

where $\mathbf{P}$ governs the dynamics of precipitation distributions, $\mathbf{X}$ is the set of dynamic sources (meridional and zonal) of the multilevel geopotential height fields, and $\mathcal{N}_T^k(\cdot)$ entails the effective nonlinear dynamic predictability associated to MDI $\mathcal{D}(\cdot)$.

The general model addresses the dynamic evolution of precipitation distributions as a function of the dynamic sources representing spatiotemporal dynamical variability in the geopotential height fields, including the embedding of the further fluid mechanical and thermodynamic footprints in the dynamics. Moreover, given the reciprocal nature of the Dynamic Interactions, it enables retroaction of Precipitation as it responds to the Dynamic Sources, conditioning their dynamics as well. This is physically important since the weather systems work towards depletion of the baroclinicity at their genesis, a negative feedback stemming from the Second Law of Thermodynamics. While the fundamental basis functionals $\mathbf{X}$ are independent, their impact on $\mathbf{P}$ as formulated in $\mathcal{N}_T^k(\cdot)$ is indeed adjusted (internal feedback) as the dynamics unfold in (26).

The dynamic model is initialised in a two-step procedure: 1) evaluating the long-term dynamical properties of the reference coevolution manifold ($\mathbf{R}$), and 2) the short-term spatiotemporal divergence of the phase space flow from $\mathbf{R}$.

- The evaluation in 1) is done via the spatial structure of the dynamic interaction between $\mathbf{P}$ and the atmospheric controls represented by the dynamic sources $\mathbf{X}$. The spatial structure plays the role of the spatial legacy of long-term dynamics as in Perdigão and Blöschl (2014).
- The procedure in 2) finds the phase space directions that maximise the equienergetic entropy production, by concentrating the Lyapunov spectrum (Ott 2002) along the highest positive values whilst conserving the spectral integral and thus the energy of the system in that initialisation step. This procedure yields similar effects to the dynamical breeding performed in (Perdigão 2010) in that it projects the dynamics onto the most unstable directions, thus maximising the entropy production of the system in line with the physical principles. However, it differs from (Perdigão 2010) in that the operation is now performed directly over the Lyapunov spectrum of the data.

Essentially, the initialisation leads to an initial distribution maximising the ability to further produce entropy in a realistic manner. In fact, while the reference manifold in 1) provides the structure of the slow dynamics (the climate or "personality" of the system), the spatiotemporal divergence in 2) provides the dominant lines of deviation from that core, i.e. the seeds for the faster weather deviations from the climate norm ("mood swings"). The dynamics then unfold as prescribed by the aforementioned model.



As an example, Figure 3.4.1 depicts a sample hindcasting experiment backward-propagating the precipitation distributions (blue shades) and pitting against real-world observations (orange line) for a randomly selected location in the Euro-Atlantic region (modelling centroid landing ca. 47.5°N, 0.5°E). In general, the observations sit within the simulated distributions, and do so in adequate quantiles (e.g. extreme events sit mostly at the corresponding extreme quantiles of the distributions).

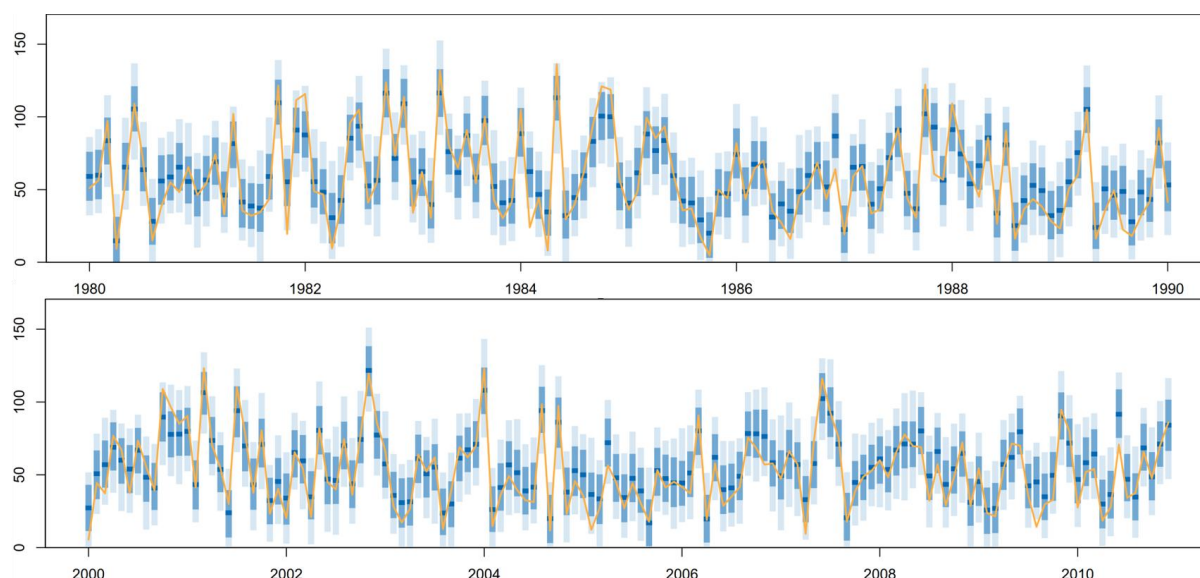


Figure 3.4.1 – Evolving precipitation distributions with quantile blue shades (lighter entailing extremes, darker closer to the median) and corresponding observational measurements (orange lines), for a randomly chosen location in the Euro-Atlantic area (modelling centroid landing ca. 47.5°N, 0.5°E).

In order to convey the spatial dimension of emerging precipitation extremes and how they respond to coevolution and synergies among underlying hydro-meteorological hazard mechanisms in geophysical dynamics (W, Φ, Λ, Θ), Figure 3.4.2 depicts the decadal spatial distribution for 2010-2020 of said response.

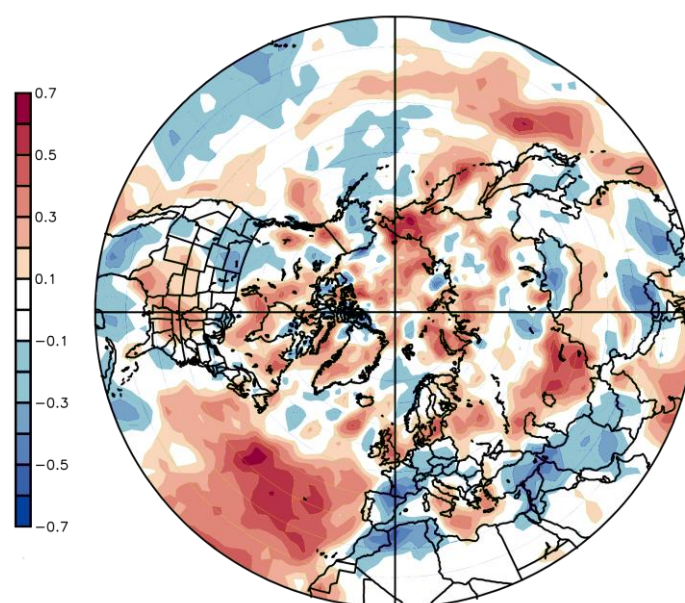


Figure 3.4.2 – Decadal (2010-2020) Geospatial Distribution of Extreme Hydro-Meteorological response to Dynamic Coevolution and Synergies among geophysical dynamic sources underneath the system compounding and cascading the multivariate processes among (W, Φ, Λ, Θ) in a thermodynamically forced setting. Negative anomalies in blue shades, positive anomalies in red shades.

The synergistic and coevolutionary metrics are dynamically derived from the MDI explained above and the overall spatiotemporal picture consistent with the polyadic synergy and redundancy metrics detailed in D2.1. While the linear and mildly nonlinear responses of precipitation extremes to ocean-atmospheric mechanisms were known, the present contribution highlights the synergistic component that cannot be attributed to individual or partial set of combined mechanisms, further stressing the need to take into account the full-fledged collective behaviour of the system dynamics at play in shaping hazardous precipitation regimes. This further highlights that the same mechanisms yield different effects in diverse locations, consistent with the nonlinearity and instability of the precipitation response insofar as strongly divergent behaviour emerges from closely aligned conditions. Sharp transitions are seen in critical areas for CSA such as Portugal, the Balearic Islands, Egaleo and Ordu, all of which lie close to the sharp interface between red and blue areas i.e. between sharply opposing signatures of precipitation response to multivariate synergies in hazard forming mechanisms. Extreme disparities in hydro-meteorological regimes are thus better understood such as the whiplash effect between extreme flood and drought conditions, along with extremely violent storm systems at play along such interfaces. These matters are being further explored in detail as the project continues to unfold.

4. Towards a Visual SoS4MHRIN Interface with Operational Results

The operational model development workflows have been accompanied with the development of a user interface through which the findings and simulations conducted herein can be better appreciated by the end-user and some relevant insights and actionable outcomes drawn for their activities.

To this end, MET's methodologies and scientific content developments have been articulated with TVS technical interface developments. The final interface is expected to be unveiled by Month 32 (MS07). Figure 4.1 below provides a sample depiction of one of the visual environments, including a list of functionalities (e.g. variables, scenarios, timeframes, spatial domains) to be included in the product.

Key Variables to be featured in the platform:

- Sea Level Pressure
- Temperature
- Wind
- Wind gust
- Precipitation
- Convective Available Potential Energy (CAPE)
- Multi-Hazard FFHMET index

CC Scenarios (for future projections):

- RCP 4.5
- RCP 8.5

Time Frames:

- Hindcasting (past)
- Nowcasting (near-present)
- Forecasting (future)

[Note: timing windows and timesteps are univocally tied to each variable, hence we only categorise by forecasting class (hind/past, forecasting/future, now/nearpresent)]

Region set (no polygons needed in fact as domain is chosen from drop-down menu):

- Global
- Euro-Atlantic
- CSA Portugal
- CSA Spain
- CSA Greece
- CSA Turkey

Lower panel tabs:

- Time Series (Ensemble vs. Observation)
- Time Series (Distribution vs. Observation)
- Information Metrics for benchmarks

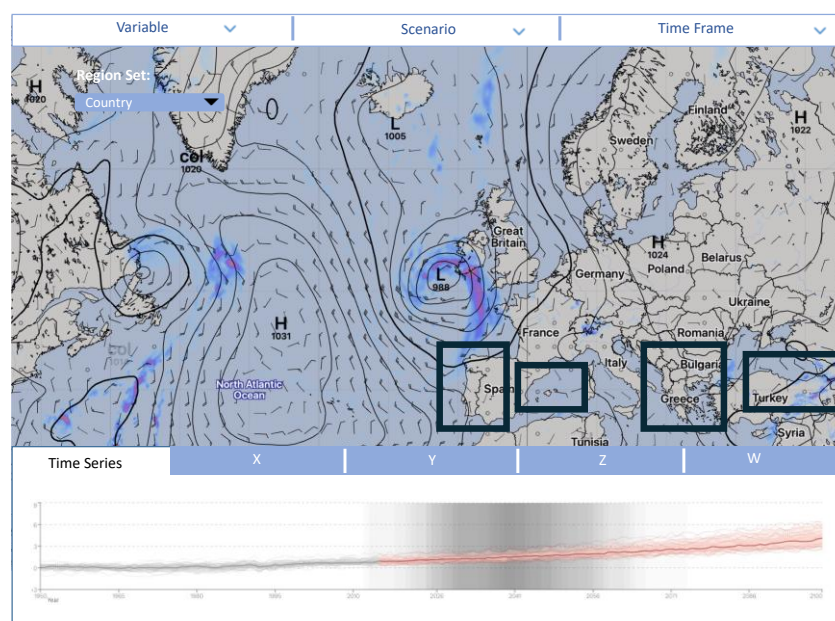


Figure 4.1 – Brief overview of key functionalities, variables, scenarios, time frames, region sets and other features being provided in the end-user visual interface prepared at TVS with MET scientific content pertaining SoSMHRIN.

In order to provide an overview to the end-user about the SoS4MHRIN methodologies and overall role at the C2IMPRESS project, the interface further provides an introduction for reference (Figure 4.2):

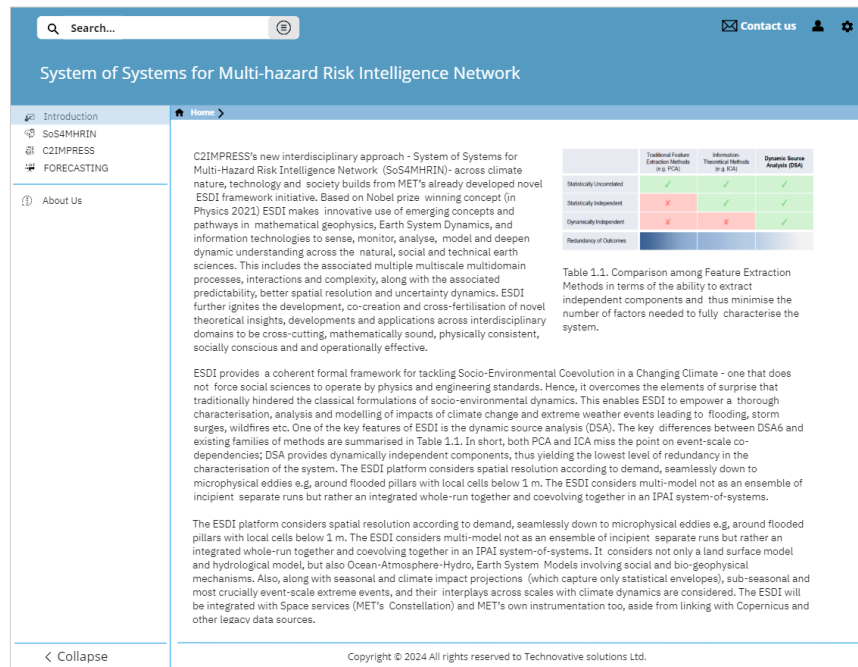


Figure 4.2 – Brief overview of the Introduction section to the SoS4MHRIN visual interface.

By entering the SoS4MHRIN service in more detail, the user finds three key features: **SoS Eye**, **SoS Analytics**, and **SoS Modelling**.

Firstly, **SoS Eye** entails the sensing or acquisition component, wherein information is retrieved from nature through MET's QITES technologies (Quantum Information Technologies in the Earth Sciences, Perdigão 2020). A sample depiction for the visual platform is provided in Figure 4.3 below:

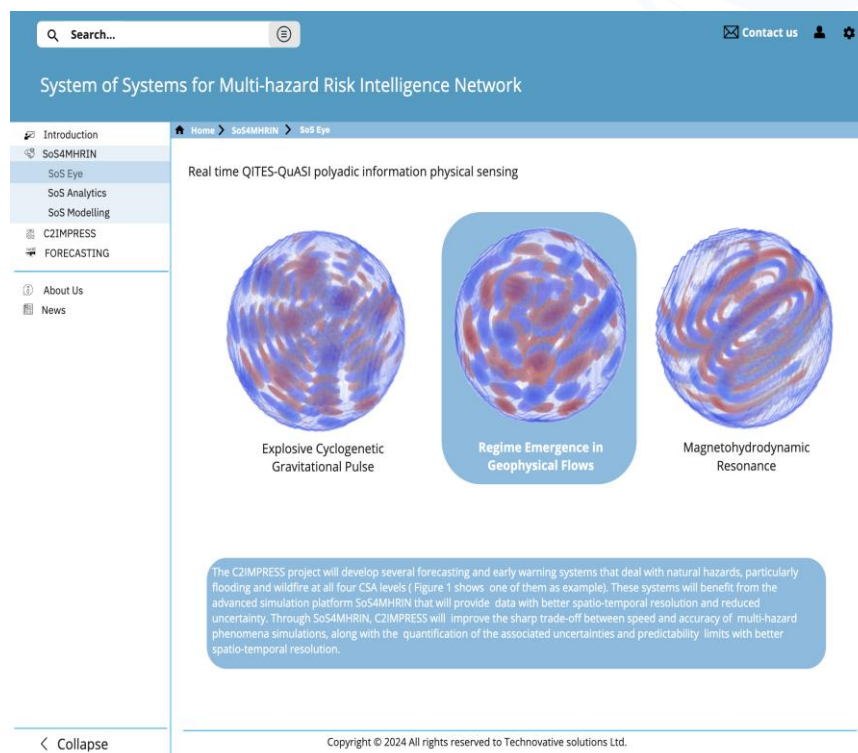


Figure 4.3 – Brief overview of the SoS Eye section to the SoS4MHRIN visual interface.

Regarding the scientific content, the figure comprises the following panels produced with MET QITES:

[Left] **Explosive Cyclogenetic Gravitational Pulse:** Information physical signature of gravitational pulse associated to the critical geopotential wave propagation arising in explosive cyclogenesis. This provides an anticipatory early warning pertaining the emergence of extreme storm systems with strong baroclinicity (pressure contrasts) and subsequent storm severity. Blue and red clusters correspond to opposite geopotential anomalies reflecting the undulatory nature of the wave propagation arising from the pulse.

[Centre] **Regime Emergence in Geophysical Flows:** Information physical early warning signature of geophysical fluid dynamical regime formation wherein flows coalesce onto vortical clusters of opposite vorticity and pressure anomalies (blue/red). An example is the formation of high- and low-pressure weather regime blocking associated to persistent events such as “blocking high” favouring dry spells, and “blocking lows” favouring the emergence of wet spells with persistent rainfall and flood-generating potential.

[Right] **Magnetohydrodynamic Resonance:** Information physical signature of magnetohydrodynamic resonance underlying critical transitions and extreme events in e.g. ionosphere and magnetosphere triggering high-level perturbations in wave-particle interactions in the atmosphere relevant to thermoelectric activity conditioning extreme storm systems.

Secondly, **SoS Analytics** entails the enhanced information physical system dynamic intelligence by MET anchored in the fundamental physical nature of information (e.g. quantum thermodynamics underneath machine learning), further augmenting it to far-from-equilibrium coevolution and synergies reflecting the systemic transformations at play in Earth System Dynamics. These include the efforts conducted in terms of dynamic source analysis to unveil non-redundant system dynamic observables underneath coevolutionary observations for dynamic model design optimization as reported in D2.1 and in the present deliverable D2.3, along with Perdigão and Hall (2023, 2024), namely Augmented Information Physical System Dynamic Intelligence for the mathematical physics procedure and Information Physical Quantum Technological Intelligence for the operationalisation.

A sample depiction of SoS Analytics output for SoS4MHRIN is portrayed in Figure 4.4 below:

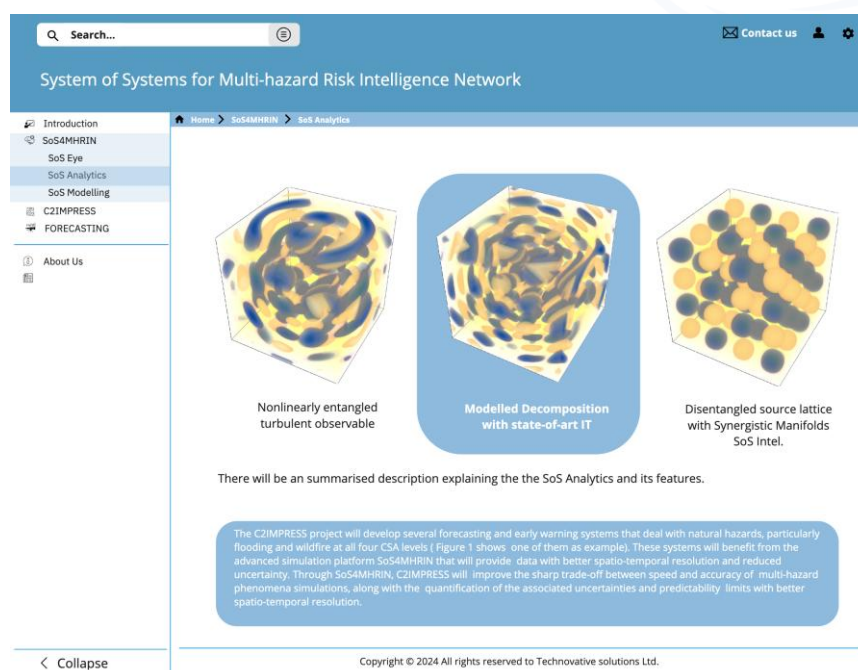


Figure 4.4 – Brief overview of the SoS Analytics section to the SoS4MHRIN visual interface.

Regarding the scientific content, the figure comprises the following panels produced with MET AIPSI:



[Left] **Nonlinearly entangled turbulent observable:** Representation of raw observational data, be it from new sensing or pre-existing dataset. Featuring a nonlinear coevolutionary mixture of interacting processes at play, e.g. turbulent mixing among fluid parcels of contrasting thermodynamic properties in oceanic (thermohaline) or atmospheric circulation.

[Centre] **Modified Decomposition with state-of-art Information Technologies:** State-of-art information technologies (IT) partially disentangle interactions to assess underlying patterns. However, such tools feature restrictive correlative links and ill-posed normality assumptions (e.g. in Principal Component Analysis) which do not hold in real-world environmental system dynamics. Therefore, PCA and related SoA machine learning feature extraction fail to extract mathematically robust physically consistent patterns, especially pertaining extreme events.

[Right] **Disentangled source lattice with Synergistic Manifolds SoS Intelligence leveraging AIPSI and IPQuTi:** Our latest generation system-of-systems intelligence, rooted on our dynamic source analysis core to ESDI, successfully disentangles the underlying physical structure, here a synergistic dynamic source lattice, the mixture of which had generated the coevolutionary turbulent mixture depicted in the left panel. This means that we can disentangle not only linear and normal processes and interactions, but also highly nonlinear, turbulent, coevolutionary and synergistic in a changing world.

Thirdly, **SoS Modelling** entails the most coveted feature of SoS4MHRIN of direct operational relevance to the end-user, namely in terms of forecasting, nowcasting and hindcasting capabilities.

In what follows, we provide one of several use cases involving a local hydrometeorologic extreme event and the underlying larger scale ocean-atmospheric context at both fluid mechanical and thermodynamic levels, showcasing the underlying multivariate multidomain interdependencies.

Predicting Ocean-Atmospheric hazard coevolution and emergence: an example

We hereby share an example use case pertaining the transition of hurricane Kirk to an extra-tropical decaying system which crossed CSA Portugal Centro during the C2IMPRESS general assembly held on October 8 and 9, 2024.

The experiment began with multiscale coevolutionary ocean-atmospheric system dynamic modelling with our ESDI core in SoS4MHRIN, conducted on October 1st at 00H UTC with lead time up to 9 days. The inputs for model initialisation are equivalent to the class of geophysical fields that routinely enter all gold-standard prediction systems in Europe (e.g. ECMWF, Deutsche Wetter Dienst, Météo-France) and North America (e.g. NOAA, NWS), including geopotential height fields at different pressure levels crucial for momentum fluxes, radiometric measurements for the energy fluxes, just to name a few. The Dynamic Sources unveiled in Perdigão (2018) and further sharpened and augmented in Perdigão and Hall (2023, 2024) encapsulate the overall system dynamic input manifold in ESDI.

In order to provide a visually intuitive overview of SoS4MHRIN capabilities in the aforementioned example, we began by predicting typical synoptic dynamics, namely in terms of dynamic proxies for pressure, wind and potential rainfall hotspots (Figure 4.5, left panel) along with wind streamfunction and gust hotspots (Figure 4.5, right panel). In doing so, the experiment showcased the ability to perform operational ocean-atmospheric predictions in line with the gold-standard prediction systems.



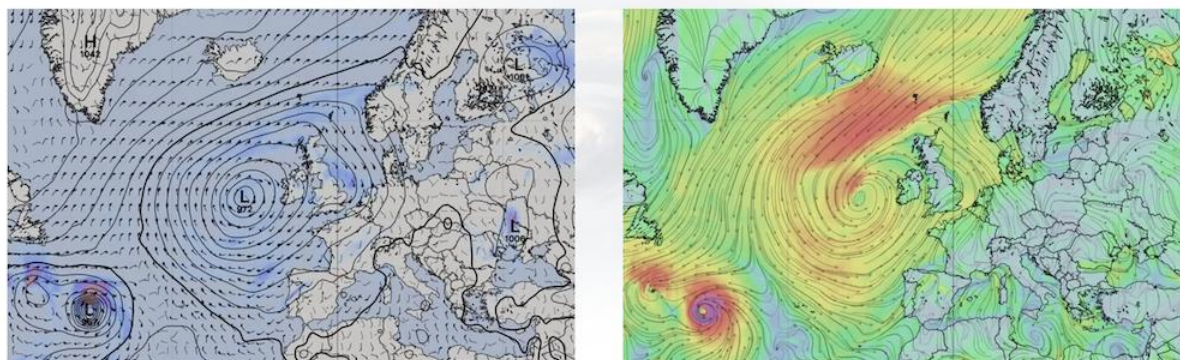


Figure 4.5 – Instantaneous snapshot of the estimated synoptic situation at 2024.10.07, 00H UTC, as prognosticated on 2024.11.01, 00H UTC. Left panel: precipitation potential risk density (coloured hotspots), sea level pressure (contours) and instantaneous wind field (wind barbs). Right panel: wind gust risk potential (coloured hotspots), and wind streamfunctions.

However, SoS4MHRIN is not only aimed to replicate gold-standards in operational ocean-atmospheric prediction systems. It aims to go beyond, capturing coevolutionary and synergistic multi-hazard risk signatures elusive to traditional gold-standard systems, with special emphasis on thermodynamic quantities that manifest themselves in nature as early warning signs of upcoming emergence of entropy production critical transitions, intermittencies, regime transitions and extremes. Therefore, our experiment proceeded by forecasting also novel multi-hazard features pertaining the emergence of high-order turbulent fluxes and geopotential carvings in the energy landscape of the synoptic ocean-atmospheric manifold, thereby providing preferential flow paths for high-order turbulent energy and moisture fluxes to coevolve and synergistically emerge as highly impactful phenomena such as convective cores, turbulent swirls and atmospheric rivers (early warning signs in Fig. 4.6).

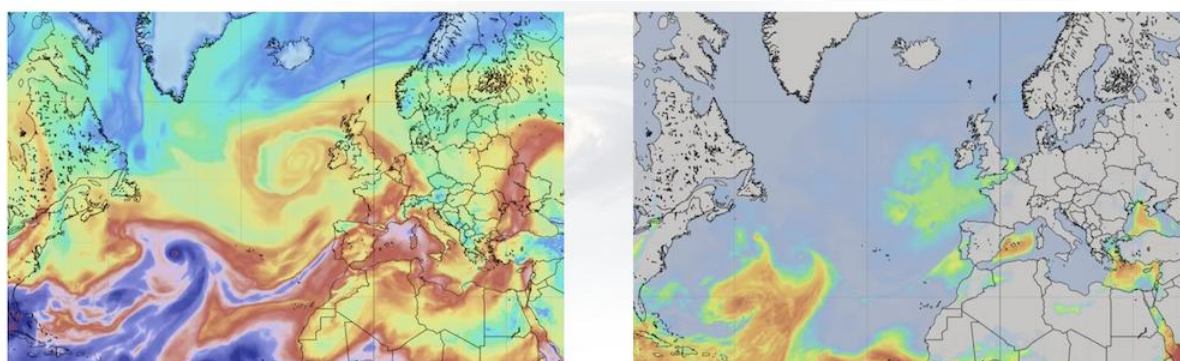


Figure 4.6 – Instantaneous snapshot of the estimated synoptic situation at 2024.10.07, 00H UTC, as prognosticated on 2024.11.01, 00H UTC and part of a dynamic run. Spatial structure of poli-hazardous multiscale spatiotemporal interplay among HyMetOc thermodynamic potentials providing early warning signatures underlying extreme events: left panel, triadic potential manifold encompassing geopotential high-order criticalities, moisture flux potential and thermodynamic turbulent kinetic energy. Right panel, dyadic among surface-air turbulent enthalpy fluxes and convective available potential energy.

Figure 4.6 demonstrates the multi-hazard nature of extreme ocean-atmospheric storm systems as emerging from polyadic coevolution and synergy among various interacting dynamic and thermodynamic mechanisms and fluxes, encompassing mass and energy fluxes, along with phase transitions exchanging mass and energy thermodynamic potentials. This is expressed in the polyadic hazard maps involving high-order gravitational wave perturbations and criticalities leading to the emergence of geopotential swirls and narrow filaments that then trigger preferential flow paths for moisture fluxes to emerge as narrow high-density atmospheric rivers, along with turbulent kinetic energy in which the motion energy is proportional not to the square of the velocity but rather to higher-order momenta, thereby expressing exponential dependence on said fluxes with

consequences in terms of high intensity, persistence and nonlinear complexity of the preferential flow paths for moisture, winds, connectivity and the overall storm synoptic outlook leading to extreme winds, rainfall and turbulence. This seeding further seeped into the Mediterranean, triggering further thermodynamic anomalies in ocean heat fluxes, atmospheric connectivity and moisture concentration, which would then feed into the cut-off systems emerging over said areas in November, an ongoing situation as this report is being written. A crucial point here is that rather than quantifying individual, cascading and compound hazard features, we encompass coevolutionary and synergistic hazard polyads in thermodynamic and geophysical fluid mechanical terms, providing further anticipatory and broader ocean-atmospheric predictive and explanatory value to impactful earth system dynamic phenomena.

SoS4MHRIN extra predictive skill in space and time

The comparison and benchmarking of SoS4MHRIN vs. state-of-art (SoA) gold-standard baselines will be reported in more detail in upcoming deliverables as the platform matures in its final form. Even so, we hereby present an early overview of some evaluation outcomes in Figure 4.7. The left panel depicts the Normalised Dynamical Information acquired in Information Physical sense as a multi-hazard subsystem termed HazFFHM performs its dynamical systems excursions at and around the Coevolutionary Earth System Dynamic (CoESD) manifold. HazFFHM entails the dynamically interactive subsystem emerging from the dynamic polyadic interplay among hydro-meteo-biogeophysical processes and feedbacks underlying cascading (time-sequential), compound (spatiotemporally collocated) and full-fledged coevolutionary emergence of fire-weather, storm-flood, heatwave-drought, and other combined hazard types as represented by a polyadic multi-hazard metric. This interplay is mathematically captured through our methodological developments reported in D2.1, namely those rooted on the Perdigão (2018) information physical metrics of polyadic entropy synergy and redundancy among dynamic processes with microphysical (e.g. event-scale) codependence, thereby reshaping the overall system dynamic behaviour and predictability, with special emphasis on eliciting elusive dependencies of crucial relevance to the emergence of multipartite extreme events.

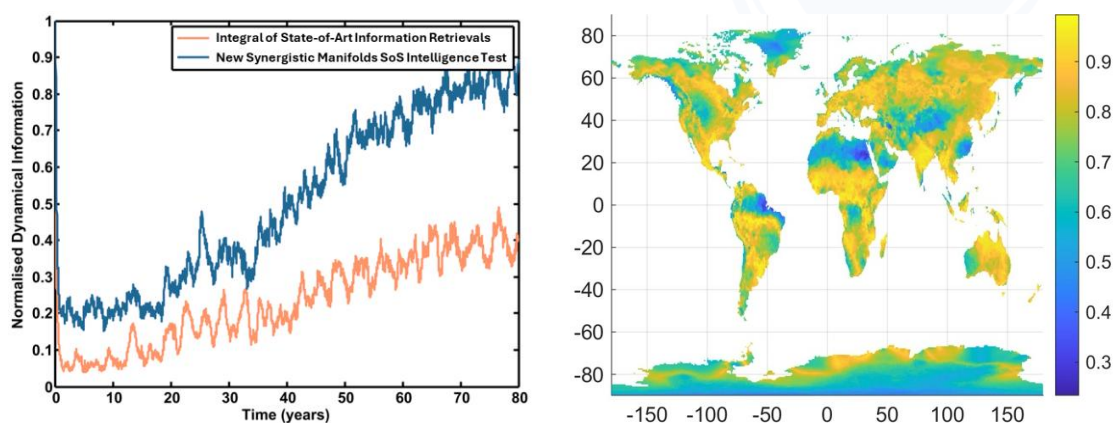


Figure 4.7 – Left panel: Multidecadal memory and predictability of coevolutionary HazFFHM orbits with substantial gains in extra skill, crucial for long-term projections. Right panel: Extra value in predictive skill of extreme HazFFHM relative to the integral of best gold-standard SoA (e.g. 0.9 means 90% increase in skill resulting in 1.9x the original baseline skill).

The integral of state-of-art information retrievals entails adding the information retrieved from gold-standard indices for each hazard type in the traditional cascading and compound approaches where the hazard information is cumulative but not synergistic. Then our edge relative to the state of the art arises from the coevolutionary and synergistic interplays that give rise to novel emerging behaviours elusive to the traditional multi-hazard approaches. This extra thus arises from the information physical metrics in ESDI (see D2.1) that consider the non-ergodic non-linear far-from-equilibrium coevolutionary and synergistic dynamic emergence associated to a changing functional-structure backbone of the dynamical system at play. Which in practice means that hazards are not just additive

as if they were a combination of separable items, but rather operate as collective multi-hazard entities. Capturing this extra information therefore provides added value in the predictability of a new dimension of multi-hazard risk intelligence, a key edge of SoS4MHRIN relative to SoA methodologies.

This extra predictability is then spatially patent in the right panel, where the scale corresponds to improvement in skill (1 being 100% increase relative to the integral of SoA information retrievals). Aggregated over seven decades (1951-2024), the mapped evaluation highlights roughly a doubling in skill over most of the continental areas, including Europe, underscoring the important of taking hazard-prone processes and interactions in a more synergistic and coevolutionary manner with our new metrics. As for areas with smaller increment in predictability, these pertain mostly deserts (North Africa, Central Asia) and other specific regions where the structural invariance of weather regimes is high enough for even the classical metrics to already fare well. That is, the highest increase in skill occurs in areas where over the years the climate system dynamics have been reshaping the CoESD attractor, disrupting its invariants and symmetries elusive to traditional methods, but captured through our ESDI core to the ongoing SoS4MHRIN framework. Further results shall be explored for other specific cases in the interface. In order to keep this deliverable succinct yet comprehensive, this example is meant to illustrate and highlight that SoS4MHRIN operates not only at one specific site, with case-specific models, but rather in a broader and more general spatiotemporal setting, thereby providing added value not only for a case-study specific level, but also in other critical areas of Europe and, with more generality, of our planet. Because C2IMPRESS aims to be not only a local solution, but also a broader solution, with an overarching general methodology, as detailed in D2.1 and here (D2.3).

5. Discussion and Conclusion

The first technological version underlying the SoS4MHRIN platform framework, termed Coevolutionary System-of-Systems Dynamic Intelligence, had been developed and reported in Deliverable 2.1. There, it was framed in the scope of our recently developed Information Physical System Dynamic Intelligence Platform leveraging MET's IPAI and ESDI methodologies and illustrated through the simulation of disaster occurrences across practical examples of this framework. Some are representative of recurrent occurrences in line with the state-of-the-art abilities of machine learning and artificial intelligence, whereas others leap beyond the SoA with the new capabilities brought up by IPAI and ESDI, harnessing and simulating unprecedented occurrences elusive to prior data records, thereby fulfilling an advance relative to the methodological state of the art.

The present deliverable (D2.3) complements the more methodological tenor of the first version with subsequent operational development, an ongoing process nearing its fruition leading to the full-fledged platform to be finally demonstrated by Month 32 (Milestone MS07).

Notwithstanding the ever-evolving nature of the operational SoS4MHRIN developments, we succinctly reported on key advances achieved thus far in said operationalisation. Firstly, the fundamental theoretical, methodological and technical developments pertaining to the System-of-Systems for Multi-Hazard Risk Intelligence Networks (SoS4MHRIN) framework were contextualised in operational model development form. Secondly, laying the scientific and methodological grounds and techniques to empower researchers, practitioners and end-users to harness diagnostic and prognostic information beyond existing data and model horizons. Thirdly, exploring emerging pathways in system dynamics beyond the possibility space spun by said past events, thereby better eliciting unprecedented features associated with the emergence of novel multi-hazard synergies. These contribute to enhancing knowledge, awareness, preparedness, predictive power and operational empowerment to tackle emerging multi-hazard challenges in a coevolutionary world.

One of the key added values of our novel ESDI implementation relative to process-based, data-based and hybrid physically informed machine learning approaches, emerges from our enhanced augmented information physical system dynamic intelligence (AIPSI). This entails the synergistic dynamic



generation of higher-dimensional geophysical dynamics from low-dimensional base fields through the dynamic source analysis explained in D2.1 and the operationalisation in the present deliverable.

Unlike classical data-based approaches that essentially identify recurrent patterns and symmetries in pre-existing data, and are thus bound to their span of representativity, our approach augments that dynamic reach through information physical intelligence, rooted on the core physical principles and formulations developed at MET as reported in deliverables D2.1, D2.3 and references therein. While a traditional physically informed machine learning approach simply injects physical equations into data-based learning workflows, our approach goes beyond by not assigning specific case-dependent equations, but rather general core principles, intrinsic to the physical infrastructure (physical intelligence), that then natively build and steer, from the physical nature underlying the computational infrastructure per se, preferential flow paths to form the system network, coevolutionary codependencies and emerging synergistic behaviours. In doing so, the system becomes no longer conditioned by a-priori choices in terms of model structure that hinder traditional platforms. This thus allows for a new window of possibilities in prognosticating system behaviours including those that had not yet manifested in past data records. Which is crucial in predicting extreme events in a changing climate through a physically consistent innovative way.

In practice, this has led to a unified framework for multiscale multidomain spatiotemporal SoS multi-hazard intelligence. Advancing system understanding and prediction of multiscale spatiotemporal compound, cascading, coevolutionary and synergistic multi-hazards. And transcended operational goals with new Coevolutionary System-of-Systems Intelligence, further enhancing SoS4MHRIN skills and functionalities relative to the pledged goals. The upcoming visual platform, to be experienced by the users in fulfilment of MS07 by M32, will provide a more palatable experience to the user pertaining these advances and functionalities in general and in use cases, given the portability and the generality of the system dynamic models across multiple spatiotemporal domains, scales and contexts.



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